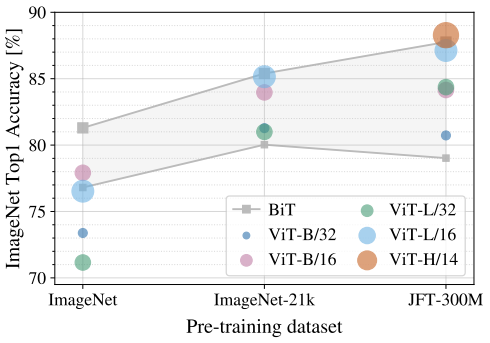

Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

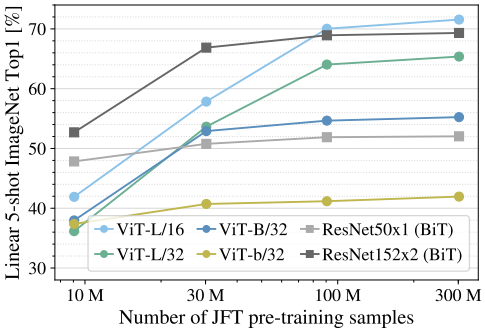

Figure 4: Linear few-shot evaluation on ImageNet versus pre-training size. ResNets perform better with smaller pre-training datasets but plateau sooner than ViT, which performs better with larger pre-training. ViT-b is ViT-B with all hidden dimensions halved.

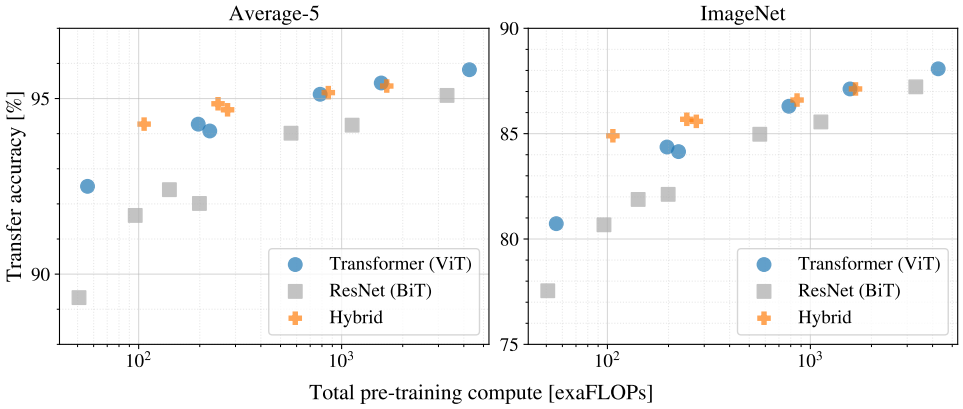

Figure 5: Performance versus pre-training compute for different architectures: Vision Transformers, ResNets, and hybrids. Vision Transformers generally outperform ResNets with the same computational budget. Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models.

region spanned by BiT models of different sizes. The BiT CNNs outperform ViT on ImageNet, but with the larger datasets, ViT overtakes.

Second, we train our models on random subsets of 9M, 30M, and 90M as well as the full JFT-300M dataset. We do not perform additional regularization on the smaller subsets and use the same hyper-parameters for all settings. This way, we assess the intrinsic model properties, and not the effect of regularization. We do, however, use early-stopping, and report the best validation accuracy achieved during training. To save compute, we report few-shot linear accuracy instead of full fine-tuning accuracy. Figure 4 contains the results. Vision Transformers overfit more than ResNets with comparable computational cost on smaller datasets. For example, ViT-B/32 is slightly faster than ResNet50; it performs much worse on the 9M subset, but better on 90M+ subsets. The same is true for ResNet152x2 and ViT-L/16. This result reinforces the intuition that the convolutional inductive bias is useful for smaller datasets, but for larger ones, learning the relevant patterns directly from data is sufficient, even beneficial.

Overall, the few-shot results on ImageNet (Figure 4), as well as the low-data results on VTAB (Table 2) seem promising for very low-data transfer. Further analysis of few-shot properties of ViT is an exciting direction of future work.

4.4 SCALING STUDY

We perform a controlled scaling study of different models by evaluating transfer performance from JFT-300M. In this setting data size does not bottleneck the models' performances, and we assess performance versus pre-training cost of each model. The model set includes: 7 ResNets, R50x1, R50x2 R101x1, R152x1, R152x2, pre-trained for 7 epochs, plus R152x2 and R200x3 pre-trained for 14 epochs; 6 Vision Transformers, ViT-B/32, B/16, L/32, L/16, pre-trained for 7 epochs, plus L/16 and H/14 pre-trained for 14 epochs; and 5 hybrids, R50+ViT-B/32, B/16, L/32, L/16 pre-trained for 7 epochs, plus R50+ViT-L/16 pre-trained for 14 epochs (for hybrids, the number at the end of the model name stands not for the patch size, but for the total dowsampling ratio in the ResNet backbone).

Figure 5 contains the transfer performance versus total pre-training compute (see Appendix D.5 for details on computational costs). Detailed results per model are provided in Table 6 in the Appendix. A few patterns can be observed. First, Vision Transformers dominate ResNets on the performance/compute trade-off. ViT uses approximately $2 - 4\times$ less compute to attain the same performance (average over 5 datasets). Second, hybrids slightly outperform ViT at small computational budgets, but the difference vanishes for larger models. This result is somewhat surprising, since one might expect convolutional local feature processing to assist ViT at any size. Third, Vision Transformers appear not to saturate within the range tried, motivating future scaling efforts.

4.5 INSPECTING VISION TRANSFORMER

To begin to understand how the Vision Transformer processes image data, we analyze its internal representations. The first layer of the Vision Transformer linearly projects the flattened patches into a lower-dimensional space (Eq. 1). Figure 7 (left) shows the top principal components of the the learned embedding filters. The components resemble plausible basis functions for a low-dimensional representation of the fine structure within each patch.

After the projection, a learned position embedding is added to the patch representations. Figure 7 (center) shows that the model learns to encode distance within the image in the similarity of position embeddings, i.e. closer patches tend to have more similar position embeddings. Further, the row-column structure appears; patches in the same row/column have similar embeddings. Finally, a sinusoidal structure is sometimes apparent for larger grids (Appendix D). That the position embeddings learn to represent 2D image topology explains why hand-crafted 2D-aware embedding variants do not yield improvements (Appendix D.4).

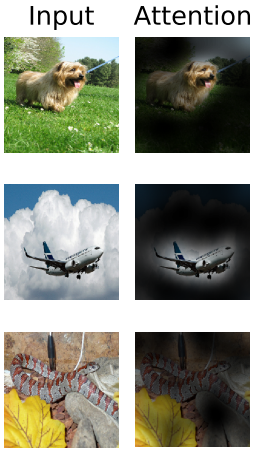

Figure 6: Representative examples of attention from the output token to the input space. See Appendix D.7 for details.

Self-attention allows ViT to integrate information across the entire image even in the lowest layers. We investigate to what degree the network makes use of this capability. Specifically, we compute the average distance in image space across which information is integrated, based on the attention weights (Figure 7, right). This "attention distance" is analogous to receptive field size in CNNs. We find that some heads attend to most of the image already in the lowest layers, showing that the ability to integrate information globally is indeed used by the model. Other attention heads have consistently small attention distances in the low layers. This highly localized attention is less pronounced in hybrid models that apply a ResNet before the Transformer (Figure 7, right), suggesting that it may serve a similar function as early convolutional layers in CNNs. Further, the attention distance increases with network depth. Globally, we find that the model attends to image regions that are semantically relevant for classification (Figure 6).

4.6 SELF-SUPERVISION

Transformers show impressive performance on NLP tasks. However, much of their success stems not only from their excellent scalability but also from large scale self-supervised pre-training (Devlin

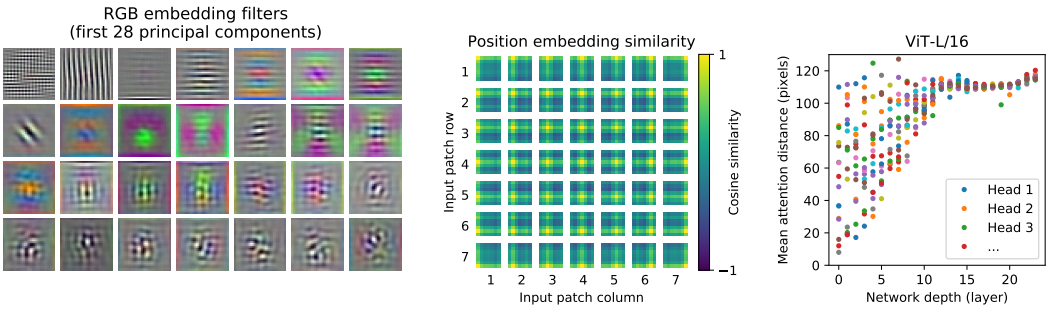

Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix D.7 for details.

et al., 2019; Radford et al., 2018). We also perform a preliminary exploration on *masked patch prediction* for self-supervision, mimicking the masked language modeling task used in BERT. With self-supervised pre-training, our smaller ViT-B/16 model achieves 79.9% accuracy on ImageNet, a significant improvement of 2% to training from scratch, but still 4% behind supervised pre-training. Appendix B.1.2 contains further details. We leave exploration of contrastive pre-training (Chen et al., 2020b; He et al., 2020; Bachman et al., 2019; Hénaff et al., 2020) to future work.

5 Conclusion

We have explored the direct application of Transformers to image recognition. Unlike prior works using self-attention in computer vision, we do not introduce image-specific inductive biases into the architecture apart from the initial patch extraction step. Instead, we interpret an image as a sequence of patches and process it by a standard Transformer encoder as used in NLP. This simple, yet scalable, strategy works surprisingly well when coupled with pre-training on large datasets. Thus, Vision Transformer matches or exceeds the state of the art on many image classification datasets, whilst being relatively cheap to pre-train.

While these initial results are encouraging, many challenges remain. One is to apply ViT to other computer vision tasks, such as detection and segmentation. Our results, coupled with those in Carion et al. (2020), indicate the promise of this approach. Another challenge is to continue exploring self-supervised pre-training methods. Our initial experiments show improvement from self-supervised pre-training, but there is still large gap between self-supervised and large-scale supervised pre-training. Finally, further scaling of ViT would likely lead to improved performance.

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